

Semisimple Algebraic Groups with No Proper Rank-One Subgroups

by

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SEMISIMPLE ALGEBRAIC GROUPS WITH NO PROPER RANK-ONE SUBGROUPS

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ABSTRACT. Let k be a number field and let v be a place of k . This thesis presents Clozel's classification, from his French-language work on Conjecture τ , of semisimple algebraic k -groups of k_v -rank 1 having no proper algebraic subgroup of k_v -rank 1. For groups of classical type, the classification shows that the only possibilities are, up to Weil restriction and isogeny, SL_2 , $\mathrm{SL}_1(D)$, and $\mathrm{SU}(D, *)$. The thesis is expository in nature: its goal is to give a detailed and accessible account of Clozel's proof, making explicit several steps that are only briefly treated in the original source.

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1. INTRODUCTION

Let k be a number field, let v be a place of k , and let k_v denote the completion of k at v . For a connected semisimple algebraic k -group G , we write $\text{rank}_{k_v} G$ for the k_v -rank of G .

The classification of algebraic groups is a central theme in the structure theory of linear algebraic groups. Beyond its intrinsic interest, it provides a framework for reducing general questions to a small number of explicit and rigid cases. In many arithmetic applications, one is led to isolate groups of small rank or with very few proper semisimple subgroups. The aim of this thesis is to study such a situation: we classify semisimple groups with k_v -rank 1 that have no proper subgroups of k_v -rank 1.

To formulate the result, we adopt Clozel's notion of groups of type $\mathcal{A}$.

Definition 1.1. *Let G be a semisimple algebraic group over k . We say that G is of type $\mathcal{A}/k$ if it is isomorphic to either*

$$(1.1) \quad \text{SL}_1(D),$$

where D is a central division algebra over k of degree d for some rational prime d , or is isomorphic to

$$(1.2) \quad \text{SU}(D, *)$$

where D is a central division algebra of degree d over some quadratic extension E/k , with d prime, and $$ is an involution on D of the second kind, i.e. acting on the center $E = Z(D)$ nontrivially.*

We say that G is of type $\mathcal{A}$ if it is k -isomorphic to the Weil restriction $\text{Res}_{k'/k} G'$, where k' is a finite extension of k , and G' is of type $\mathcal{A}/k'$.

We also recall that a connected reductive group G is of *classical type* if its Dynkin diagram is of type A , B , C , or D .

The main theorem of this thesis is the following result of Clozel.

Theorem 1.2 ([Clo03, Thm. 1.1]).

- (1) *Let G be an anisotropic, semisimple, and absolutely simple algebraic group over k . If G is of classical type, then there exists a algebraic subgroup $H \subseteq G$ defined over k , which is isogenous to a group of type $\mathcal{A}$.*
- (2) *In addition, if G is simply connected, then H may be chosen to be simply connected, and hence itself of type $\mathcal{A}$.*
- (3) *Let v be a place of k , and k_v be the v -adic completion. If the k_v -rank of G is equal to 1, then one can choose a subgroup $H \subseteq G$ over k such that H also has k_v -rank 1. If G is simply connected, then H may again be taken to be simply connected.*

This theorem immediately restricts the possible semisimple k -groups with $\text{rank}_{k_v} G = 1$ and with no proper subgroup of k_v -rank 1. Indeed, suppose G admits no proper algebraic subgroup $H \subsetneq G$ with $\text{rank}_{k_v} H = 1$. If G is anisotropic over k , then by Clozel's theorem any such group of classical type contains a k -subgroup of type $\mathcal{A}$ who k_v -rank is still 1. Therefore, unless this subgroup is the whole group, we obtain a contradiction. It follows that such a group G must itself already belong to the rank-1 cases occurring in type $\mathcal{A}$, together with the split rank-1 case SL_2 . Concretely, one is reduced to the groups

$$\text{SL}_2, \quad \text{SL}_1(D), \quad \text{SU}(D, *).$$

The proof presented in this thesis follows Clozel's remarkable work in [Clo03, § 1], in which he proves the Conjecture τ . This classification plays an important role in the proof of the conjecture. The point is that Kazhdan's property (T) is already known for groups of rank at least 2, so one is reduced to the case of groups of k_v -rank 1. Reduction principles in [BS91] and [CU04] then allows

one to restrict further to semisimple groups having no proper subgroup of rank 1. Classifying such groups is therefore a decisive structural step: it shows that only the explicit groups SL_2 , $\mathrm{SL}_1(D)$, and $\mathrm{SU}(D, *)$ remain, and these can then be treated individually.

The thesis is organized as follows. Section 2 reviews the preliminary material required for the proof of the main theorem, including first Galois cohomology and root data. Section 3 contains the proof of the main theorem, carried out by a case-by-case analysis according to the classical Dynkin types A , B , C , and D .

2. PRELIMINARIES

2.1. Algebraic Groups.

Throughout this paper, a *group variety* means a group object in the category of abstract algebraic varieties, and an *algebraic group* means a smooth group variety. We refer the reader to [Vak25, Ch. 11] for more precise definitions. Unless stated otherwise, all algebraic groups considered in this paper are linear; in particular, they are affine from the scheme-theoretic point of view.

Let K be an algebraically closed field, with subfields $F \subseteq E \subseteq K$. Let G be an algebraic group over E . We say G is an F -group if there exists an algebraic group G_F over F such that $G \cong G_F \times_{\mathrm{Spec} F} \mathrm{Spec} E$. The group G_F is called an F -structure of G . For brevity, we will sometimes write $G = G_F \times_F E$.

Now assume that G is an F -group with fixed F -structure G_F . A subset $U \subseteq G$ is said to be F -closed (resp. F -open) if it is closed (resp. open) for the topology on G induced by the projection $\pi : G \rightarrow G_F$. If $U \subseteq G$ is an F -open subset, we write $F[U]$ for the ring of regular functions on the corresponding open subset in G_F . If G and G' are both F -groups over E , then a morphism of algebraic groups $f : G \rightarrow G'$ is called an F -morphism if it is compatible with the given F -structures.

Take L to be an extension of E . For open subset $U \subseteq G$, we write $L[U]$ for the ring of regular functions on the corresponding open subset of the base change $G_L = G \times_{\mathrm{Spec} E} \mathrm{Spec} L$. An L -rational point g of G is a E -scheme morphism $g : \mathrm{Spec} L \rightarrow G$. We write

$$G(L) := \mathrm{Hom}_E(\mathrm{Spec} L, G)$$

for the set of all L -rational points of G . At times, we will also identify such a morphism g with its corresponding scheme-theoretic point, together with the datum that its residue field is L . We will pass between these two viewpoints without further comment.

In addition to the preceding algebro-geometric notions, we assume familiarity with the definitions of semisimple groups, reductive groups, and (absolutely) simple groups, as well as the basic theory surrounding them; standard references are [Hum75] and [Bor91]. We also use the notions of a torus and an F -split torus, which lead to the definition of the F -rank of an algebraic group G ; see [Mar91, § 0.22].

2.2. Galois Cohomology.

Let Γ be a profinite group, and let A be a group equipped with the discrete topology. Suppose that Γ acts continuously on A . A 1-cocycle of Γ in A is a continuous function $c : \Gamma \rightarrow A$ such that for any $\gamma, \delta \in \Gamma$, we have $c(\gamma\delta) = c(\gamma)(\gamma.c(\delta))$. The collection of all such cocycles is denoted by $Z^1(\Gamma, A)$. We define an equivalence relation $\sim$ on $Z^1(\Gamma, A)$ by declaring $c \sim d$ if there exists $a \in A$ such that $d(\gamma) = a^{-1}c(\gamma)(\gamma.a)$ for all $\gamma \in \Gamma$. The 1-cohomology set of Γ in A is then defined by

$$H^1(\Gamma, A) := Z^1(\Gamma, A) / \sim .$$

The set $H^1(\Gamma, A)$ is naturally a pointed set, whose distinguished point is the equivalence class of the trivial cocycle $c_0 : \gamma \rightarrow 1$. If A is abelian, then $H^1(\Gamma, A)$ is in fact an abelian group, with group law induced by pointwise multiplication of cocycles.

Now, let F be a subfield of an algebraically closed field K , and let $F_s \subseteq K$ be a separable closure of F . Write $\Gamma_s = \text{Gal}(F_s/F)$. Let G be an algebraic group over F , and let R denote its coordinate ring. Then $F_s[G] = R \otimes_F F_s$, and Γ_s acts on $F_s[G]$ by $\gamma.(r \otimes s) := r \otimes \gamma(s)$.

Proposition 2.1. *The action of Γ_s on $F_s[G]$ is continuous relative to the Krull topology on Γ_s and the discrete topology on $F_s[G]$.*

Proof. Since $F_s[G]$ is equipped with the discrete topology, it suffices to show that for every $f = \sum_1^n r_i \otimes s_i \in F_s[G]$, the stabilizer $\text{Stab}_{\Gamma_s}(f)$ is open. Let $E = F(s_1, \dots, s_n)$, and choose a finite Galois extension E'/F contained in F_s such that $E \subseteq E'$. Then for every $\sigma \in \text{Gal}(F_s/E')$, we have $\sigma(s_i) = s_i$ for all $1 \leq i \leq n$. Hence

$$\sigma.f = \sum_{i=1}^n r_i \otimes \sigma(s_i) = f.$$

Therefore, $\text{Gal}(F_s/E')$ is an open subgroup of $\text{Stab}_{\Gamma}(f)$. It follows that $\text{Stab}_{\Gamma_s}(f)$ is open. This proves the continuity of the action. $\square$

Since G is affine, there is a natural bijection between the F_s -automorphisms of the F_s -algebra $F_s[G]$ and the F_s -automorphisms of the algebraic group G_{F_s} . Thus, the preceding action of Γ_s induces an action of Γ_s on G_{F_s} . Endowed with the discrete topology, G_{F_s} becomes a continuous Γ_s -set by the same argument as in Proposition 2.1. Moreover, each $\sigma \in \Gamma_s$, viewed as an automorphism of G_{F_s} , preserves residue fields up to canonical isomorphism:

$$\kappa(g) \cong \kappa(\sigma(g)) \quad \text{for all } g \in G_{F_s}.$$

In particular, σ sends F_s -rational points to F_s -rational points, and hence induces an action of Γ_s on $G(F_s)$. With respect to this action, we may form the 1-cocycle set $Z^1(\Gamma_s, G(F_s))$ and 1-cohomology set $H^1(\Gamma_s, G(F_s))$. In most cases, we abbreviate these as $Z^1(F, G)$ and $H^1(F, G)$, respectively, and refer to $H^1(F, G)$ as *the first Galois cohomology set* of G .

Let $\text{Aut}_{F_s}(G)$ denote the group of F_s -automorphisms of the algebraic group G_{F_s} . The action of Γ_s on G_{F_s} induces, by conjugation, an action on $\text{Aut}_{F_s}(G)$. This action is also continuous with respect to the discrete topology on $\text{Aut}_{F_s}(G)$, since the stabilizer of each element is an open subgroup of Γ_s . Consequently, we may form the 1-cohomology set $H^1(\Gamma_s, \text{Aut}_{F_s}(G))$.

We define an F -form of G to be an algebraic group G' over F such that G'_{F_s} is F_s -isomorphic to G_{F_s} . Let $\Phi(F, G)$ denote the set of F -isomorphic classes of F -forms of G . There is a close relationship between F -forms of G and 1-cocycles of Γ_s with values in $\text{Aut}_{F_s}(G)$. Indeed, Let G' be an F -form of G , and choose an F_s -isomorphism $\phi : G'_{F_s} \rightarrow G_{F_s}$. Then we define a map c_ϕ by

$$c_\phi(\gamma) := \phi \circ \gamma \circ \phi^{-1} \circ \gamma^{-1}.$$

It is straightforward to verify that c_ϕ is a 1-cocycle, and that its equivalence class is independent of the choice of the isomorphism ϕ . Furthermore, the assignment $G' \mapsto c_\phi$ descends to a well-defined map $\Phi(F, G) \rightarrow H^1(\Gamma_s, \text{Aut}_{F_s}(G))$, which leads to the following canonical result.

Proposition 2.2 ([Spr98, Prop. 11.3.3]). *The above construction defines a bijection $\Phi(F, G) \rightarrow H^1(\Gamma_s, \text{Aut}_{F_s}(G))$, which sends the class of G to the distinguished class of 1.*

Now recall that each point $g \in G(F_s)$ defines an inner automorphism $\text{Int}_g : G_{F_s} \rightarrow G_{F_s}$ by $\text{Int}_g(x) = gxg^{-1}$. Thus, we obtain a group homomorphism

$$\text{Int} : G(F_s) \rightarrow \text{Aut}_{F_s}(G), \quad g \mapsto \text{Int}_g.$$

We now describe how this homomorphism induces a map of pointed sets $H^1(F, G) \rightarrow \Phi(F, G)$. Since Int is Γ_s -equivariant, it induces a map $H^1(F, G) \rightarrow H^1(\Gamma_s, \text{Aut}_{F_s}(G))$. More explicitly, the class of a cocycle $c : \Gamma_s \rightarrow G(F_s)$ is sent to the class of cocycle $\text{Int} \circ c$. Composing this with the inverse of the bijection in Proposition 2.2, we obtain a map

$$H^1(F, G) \rightarrow H^1(\Gamma_s, \text{Aut}_{F_s}(G)) \rightarrow \Phi(F, G).$$

More precisely, the class of cocycle c is mapped to the F -form $G_c := \text{Spec } F[G]_c$, where the F -algebra $F[G]_c$ is defined by

$$F[G]_c := \{f \in F_s[G] : \text{Int} \circ c(\gamma)(\gamma.f) = f \text{ for all } \gamma \in \Gamma_s\}.$$

By [Spr98, Prop. 11.1.6], the algebraic group G_c over F is indeed an F -form of G .

Definition 2.3. *An F -form of G whose F -isomorphism class lies in the image of $H^1(F, G) \rightarrow \Phi(F, G)$ is called an inner form of G . Otherwise it is called an outer form of G .*

2.3. Root Datum.

In this section, we introduce the language of root data, a combinatorial framework for the studying the structure of connected reductive algebraic groups. In view of the length of this thesis, we present only the definitions and results that are needed later, omitting most proofs. More detailed accounts of this subject may be found in [Spr98, Chs. 7, 8, and 17] and [Mil22, Chs. 21, 23, and Appendix C].

The notion of a root datum originates in SGA 3 [DG70, Ch. XXI].

Definition 2.4. *A root datum is a quadruple $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$, where X and $X^\vee$ are free $\mathbb{Z}$ -modules of finite rank in duality by a pairing $\langle \cdot, \cdot \rangle : X \times X^\vee \rightarrow \mathbb{Z}$, and $\Phi \subseteq X$, $\Phi^\vee \subseteq X^\vee$ are finite subsets equipped with a bijection $\Phi \rightarrow \Phi^\vee$, written $\alpha \mapsto \alpha^\vee$.*

The root datum Ψ is required to satisfy the following two axioms:

RD1 *For every $\alpha \in \Phi$, we have $\langle \alpha, \alpha^\vee \rangle = 2$.*

RD2 *For each $\alpha \in \Phi$, we have $s_\alpha(\Phi) \subseteq \Phi$ and $s_{\alpha^\vee}(\Phi^\vee) \subseteq \Phi^\vee$, where the maps $s_\alpha : X \rightarrow X$ and $s_{\alpha^\vee} : X^\vee \rightarrow X^\vee$ are defined by*

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha, \quad s_{\alpha^\vee}(y) = y - \langle \alpha, y \rangle \alpha^\vee.$$

The elements of Φ and $\Phi^\vee$ are called the *roots* and *coroots* of Ψ , respectively; for each $\alpha \in \Phi$, the corresponding $\alpha^\vee$ is called the coroot associated with α . Axiom RD1 implies $s_\alpha(\alpha) = -\alpha$ and $s_{\alpha^\vee}(\alpha^\vee) = -\alpha^\vee$. Therefore, it is natural to refer to s_α and $s_{\alpha^\vee}$ as the *reflections* of α and $\alpha^\vee$. The subgroup of the automorphism group $\text{Aut}(X)$ generated by $\{s_\alpha : \alpha \in \Phi\}$ is called the *Weyl group* of Ψ , and is denoted by $W(\Psi)$. The Weyl group is finite, since it acts faithfully on the finite set Φ .

It is straightforward to verify that if $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ is a root datum, then Φ is a *root system* in the real vector space $V := \mathbb{Z}\Phi \otimes_{\mathbb{Z}} \mathbb{R}$. Here, “root system” is understood in the classical sense; see [Mar91, § 0.26]. Notice that each reflection s_α extends $\mathbb{R}$ -linearly to an automorphism of the vector space V , and we again write $W(\Psi)$ for the subgroup of $\text{Aut}(V)$ generated by these reflections. Choose any positive-definite inner product $(\ , \)_0$ on V . Since $W(\Psi)$ is finite, averaging over the group yields a $W(\Psi)$ -invariant positive-definite inner product

$$(v, w) := \sum_{g \in W(\Psi)} (gv, gw)_0.$$

With respect to this inner product, each s_α is an orthogonal reflection across the hyperplane $H_\alpha = \{v \in V : (v, \alpha) = 0\}$. Those reflecting hyperplanes cut the space V into connected components, and each connected component of $V \setminus \bigcup_{\alpha \in R} H_\alpha$ is called a *Weyl chamber*.

After choosing a Weyl chamber C , we obtain a notion of positivity. A root $\alpha \in R$ is called *positive* if $(v, \alpha) > 0$ for all $v \in C$. The remaining roots are called *negative*. A positive root that cannot be expressed as the sum of two positive roots are called a *simple root*. We denote the set of simple roots associated with C by $\Delta(C) = \{\alpha_1, \dots, \alpha_r\}$. We can check that $\Delta(C)$ spans the whole space V . It can also be verified that if $\alpha_i, \alpha_j \in \Delta(C)$ are distinct

simple roots, then, after possibly interchanging i and j , their pairings fall into exactly one of the following four cases:

- (1) $\langle \alpha_i, \alpha_j^\vee \rangle = \langle \alpha_j, \alpha_i^\vee \rangle = 0$;
- (2) $\langle \alpha_i, \alpha_j^\vee \rangle = \langle \alpha_j, \alpha_i^\vee \rangle = -1$;
- (3) $\langle \alpha_i, \alpha_j^\vee \rangle = -2, \langle \alpha_j, \alpha_i^\vee \rangle = -1$;
- (4) $\langle \alpha_i, \alpha_j^\vee \rangle = -3, \langle \alpha_j, \alpha_i^\vee \rangle = -1$.

The *Dynkin diagram* is a directed graph whose vertices correspond to the simple roots. For two distinct vertices α_i and α_j , the number of edges between them, together with the direction of the arrow, is determined by which of the four cases above occurs, as illustrated in the following figure.

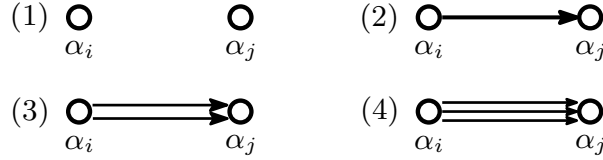


FIGURE 1. Four Edge Relations

Since $W(\Psi)$ acts transitively on the set of Weyl chambers, the Dynkin diagram is independent of the choice of chamber C . We say that a root datum $\mathbb{R}$ is *irreducible* if its Dynkin diagram is connected. It can be shown that $\mathbb{R}$ is irreducible if and only if its corresponding root system is not a disjoint union of two nonempty subsystems. Dynkin's fundamental work [Dyn47] shows that every connected Dynkin diagram belongs to one of seven families in the figure below. This yields, in particular, a complete classification of irreducible root data.

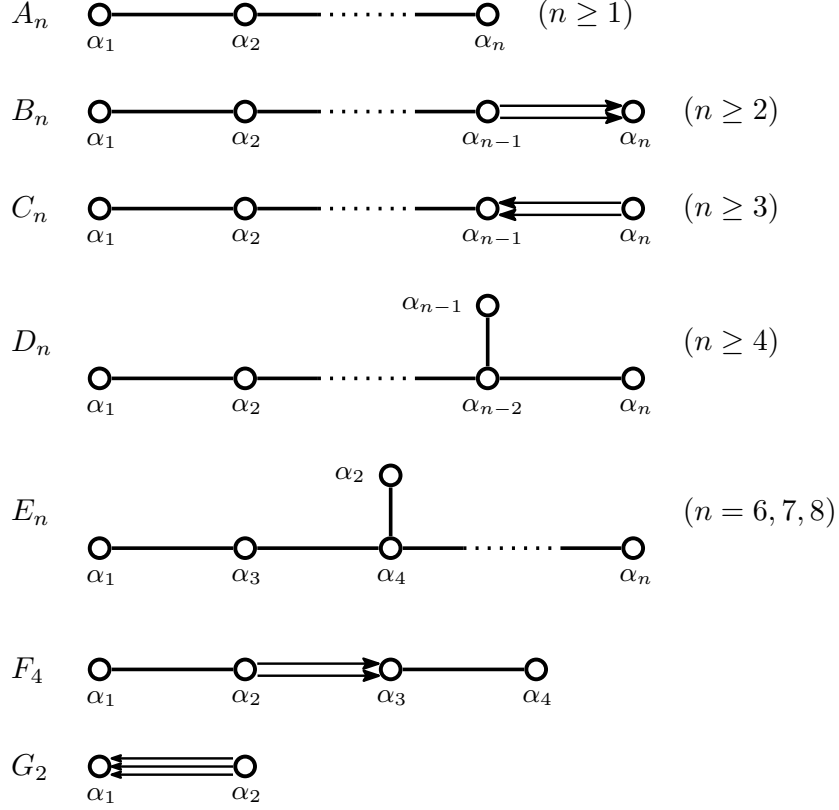


FIGURE 2. Classification of Connected Dynkin Diagrams

We now explain how to associate a root datum to an algebraic group. Let G be a connected reductive algebraic group over a field F , and let $T \subseteq G$ be a maximal torus that is split over F . For brevity in later texts, we call such a pair (G, T) a *connected split reductive group*. We define the *character* and *cocharacter* groups of T by

$$X(T) := \text{Hom}(T, \mathbb{G}_m), \text{ and } X^\vee(T) := \text{Hom}(\mathbb{G}_m, T),$$

respectively. These are finitely generated free abelian groups, and they are in perfect duality via the pairing $\langle \chi, \lambda \rangle \in \mathbb{Z}$, defined by the identity $\chi \circ \lambda(t) = t^{\langle \chi, \lambda \rangle}$ for all $t \in \mathbb{G}_m$.

To define the roots, consider the Lie algebra $\mathfrak{g} = \text{Lie}(G)$. The conjugation action of T on G induces, by differentiation, the *adjoint representation* $\text{Ad} : T \rightarrow \text{GL}(\mathfrak{g})$. Since T is a split torus, this representation is diagonalizable.

Accordingly, $\mathfrak{g}$ admits a weight-space decomposition

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\chi \in X(T) \setminus \{0\}} \mathfrak{g}_\chi,$$

where for each $\chi \in X(T)$,

$$\mathfrak{g}_\chi := \{X \in \mathfrak{g} : \text{Ad}(t)(X) = \chi(t)X \text{ for all } t \in T\}.$$

The set of *roots of G with respect to T* is then defined by

$$\Phi(G, T) := \{\chi \in X(T) \setminus \{0\} : \mathfrak{g}_\chi \neq 0\}.$$

Since T is maximal, the zero-weight space $\mathfrak{g}_0$ is precisely $\mathfrak{t} := \text{Lie}(T)$. Hence the above decomposition may be rewritten as

$$(2.1) \quad \mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\chi \in \Phi(G, T)} \mathfrak{g}_\chi.$$

Now, we define the Weyl group of (G, T) to be the étale group scheme

$$W(G, T) = N_G(T)/Z_G(T).$$

Let $w \in W(G, T)$ be represented by an element $n \in N_G(T)$. Since n normalizes T , conjugation by n induces an automorphism of T . Consequently, $W(G, T)$ acts on $X(T)$ by

$$(w.\chi)(t) := \chi(n^{-1}tn), \quad \chi \in X(T), t \in T.$$

By the discussion above, we may choose a $W(G, T)$ -invariant positive-definite inner product on the real vector space spanned by $\Phi(G, T)$, and for each $\alpha \in \Phi(G, T)$ let s_α denote the corresponding orthogonal reflection. Then, by [Spr98, Lems. 7.1.8], for each $\alpha \in \Phi(G, T)$ there exists a cocharacter $\alpha^\vee \in X^\vee(T)$ satisfies axioms RD1 and RD2. We denote by $\alpha^\vee$'s by $\Phi^\vee(G, T)$ the set of such coroots. This leads to the following definition.

Definition 2.5. *Let (G, T) be a connected split reductive group over F . The root datum of G with respect to T is defined to be the quadruple*

$$\Psi(G, T) = \{X(T), \Phi(G, T), X^\vee(T), \Phi^\vee(G, T)\}.$$

Moreover, one verifies that the Weyl group associated with $\Psi(G, T)$ identifies with the subgroup of $\text{Aut}(X(T))$ generated by the reflections s_α for $\alpha \in \Phi(G, T)$ (see, e.g., [Spr98, Thm. 7.1.9]).

We now introduce several fundamental results that illustrate the central role of root data in the theory of reductive groups. To state these results, we first need a few additional definitions.

Definition 2.6. *Let $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ be a root datum. We say Ψ is reduced if for every $\alpha \in \Phi$, we have $\mathbb{Q}\alpha \cap \Phi = \{\alpha, -\alpha\}$.*

Definition 2.7. *Let $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ and $\Psi' = (X', \Phi', X'^\vee, \Phi'^\vee)$ be root data. An isomorphism of root data $\Psi' \rightarrow \Psi$ is a group isomorphism $f : X' \rightarrow X$ such that $f(\Phi') = \Phi$ and $f^\vee(\Phi'^\vee) = \Phi^\vee$.*

Now assume that Ψ and Ψ' are the root data of connected reductive groups (G, T) and (G', T') over a field F , respectively. Let p denote the characteristic exponent of F . Then an isogeny of root data $\Psi' \rightarrow \Psi$ is an injective group homomorphism $f : X' \rightarrow X$ with finite cokernel, together with a bijection $\cdot' : \Phi \rightarrow \Phi'$ and a set map $q : \Phi \rightarrow p^\mathbb{N}$ such that $f(\alpha') = q(\alpha)\alpha$ and $f^\vee(\alpha'^\vee) = q(\alpha)\alpha^\vee$ for every $\alpha \in \Phi$.

Theorem 2.8 (Existence Theorem, [Mil22, Thm. 23.55]). *Let $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ be a reduced root datum. Then there exists a connected split reductive algebraic group G , together with a split maximal torus $T \subseteq G$ such that $\Phi(G, T) \cong \Psi$.*

Theorem 2.9 (Isomorphism/Isogeny Theorem, [Mil22, Thm. 23.25]). *Let (G, T) and (G', T') be connected split reductive algebraic groups. Then every*

isomorphism of root data $\Psi(G, T) \rightarrow \Psi(G', T')$ is induced by an isomorphism of algebraic groups $G \rightarrow G'$ carrying T to T' .

The statement remains valid with “isomorphism” replaced throughout by “isogeny”.

Two important observations can be made using the theorems above. First, let $T_1, T_2 \subseteq G$ be two split maximal tori. By [Mil22, Thm. 17.105], there exists an element $g \in G(F)$ such that $T_1 = \text{inn}(g)T_2$. The inner automorphism $\text{inn}(g) : (G, T_2) \rightarrow (G, T_1)$ then induces an isomorphism of root data $\Psi(G, T_2) \rightarrow \Psi(G, T_1)$. Hence, for a split reductive group G , the isomorphism class of the root datum depends only on G , and not on the choice of split maximal torus T .

Second, it is immediate from the definition that the root datum attached to a split reductive group is unchanged under extension of the base field. Consequently, if G is an arbitrary connected reductive group over F and $T \subseteq G$ is a maximal torus, then after base change to a separable closure F_s , the torus T_{F_s} becomes split, and we may consider the root datum $\Psi(G_{F_s}, T_{F_s})$. This root datum is called the *absolute root datum* of G .

Theorem 2.9 shows that an isomorphism of root data gives rise to an isomorphism of the corresponding connected split reductive groups. Although such an isomorphism of algebraic groups need not be unique, one can make a canonical choice by specifying a distinguished system of positive roots via Borel subgroups. A *Borel subgroup* of an algebraic group G is a maximal connected solvable closed subgroup. Since every maximal torus of G is connected and solvable, it is contained in a Borel subgroup B . Such a pair (B, T) is called a *Borel pair* of G .

Now assume in addition that G is connected and reductive, and fix a maximal torus $T \subseteq B$. Then, in the root space decomposition (2.1), we define

$$\Phi^+(G, B, T) = \{\alpha \in \Phi(G, T) : \mathfrak{g}_\alpha \subseteq \mathfrak{b}\},$$

where $\mathfrak{b} = \text{Lie}(B)$. By [Mil22, pp. 435], this is a system of positive roots, and we denote the associated set of simple roots by $\Delta(G, B, T)$.

We next define a *based root datum* to be a root datum $(X, \Phi, X^\vee, \Phi^\vee)$ together with a choice of simple roots $\Delta \subseteq \Phi$. An isomorphism (resp. isogeny) of based root data is an isomorphism (resp. isogeny) of root data that sends simple roots to simple roots. A *pinned reductive group* is a connected split reductive group (G, T) together with a pair $(\Delta, (e_\alpha)_{\alpha \in \Delta})$, where $\Delta = \Delta(G, T)$ is a choice of simple roots in its root datum, and e_α is a nonzero element of $\mathfrak{g}_\alpha$ (in the sense of (2.1)) for each $\alpha \in \Delta$. An isomorphism (resp. isogeny)

$$f : (G, T, \Delta, (e_\alpha)) \rightarrow (G', T', \Delta', (e_{\alpha'}))$$

of pinned reductive groups is an isomorphism (resp. isogeny) of reductive groups that sends each simple root $\alpha \in \Delta$ to a simple root $\alpha' \in \Delta'$, and satisfies $f(\exp(e_\alpha)) = \exp(e_{\alpha'})$ for every $\alpha \in \Delta$. With this terminology in place, we can now state the “uniqueness” version of Theorem 2.9.

Theorem 2.10 (Fundamental Theorem, [Mil22, Thm. 23.40]). *Let G and G' be pinned reductive groups, and let $f : \Psi(G) \rightarrow \Psi(G')$ be an isomorphism of the corresponding based root data. Then there exists a unique isomorphism $\phi : G \rightarrow G'$ of pinned reductive groups inducing f .*

The statement remains valid with “isomorphism” replaced throughout by “isogeny”.

By combining Theorem 2.10 with the construction of the Borel pair (B, T) and the associated set of simple roots $\Delta(G, B, T)$, we obtain a canonical choice

of isomorphism of reductive groups arising from an isomorphism of root data. This construction will also play an essential role in Section 3.1.2, where we introduce the Langlands dual group.

The preceding results clarify the bijective correspondence, up to isomorphism, between split reductive algebraic groups and root data. One should note, however, that a single root system may correspond to more than one root datum, so in general algebraic groups cannot be classified solely by their root systems or Dynkin diagrams. This ambiguity disappears in the semisimple case.

For a root datum $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$, we say that Ψ is *semisimple* if Φ spans the real vector space $X \otimes_{\mathbb{Z}} \mathbb{R}$. As shown in [Mil22, Prop. 21.48], a split reductive group is semisimple if and only if its root datum is semisimple, which justifies the terminology. For a semisimple root datum Ψ , we also define its *root lattice* $Q(\Phi)$ and *weight lattice* $P(\Phi)$ by

$$Q(\Phi) = \mathbb{Z}\Phi, \quad P(\Phi) = \{\lambda \in \Phi \otimes_{\mathbb{Z}} \mathbb{R} : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \text{ for all } \alpha \in \Phi\},$$

and these satisfy $Q(\Phi) \subseteq X \subseteq P(\Phi)$ by [Mil22, Prop. C.35]. We observe that these two lattices are determined solely by the root system Φ , and in particular do not depend on the choice of the intermediate lattice X . Therefore, to construct a semisimple root datum from a given root system $(\Phi \otimes_{\mathbb{Z}} \mathbb{R}, \Phi)$, it is equivalent to choose a lattice X that lies between $Q(\Phi)$ and $P(\Phi)$. This leads to the following theorem.

Theorem 2.11 ([Mil22, Thm. 23.58]). *There is a bijection between the set of isomorphism classes of split semisimple algebraic groups and the set of isomorphism classes of triples (V, Φ, X) , where (V, Φ) is a root system and X is a lattice satisfying $Q(\Phi) \subseteq X \subseteq P(\Phi)$.*

For a simply connected split semisimple group (G, T) , we always have $X(G, T) = P(\Phi(G, T))$ by [Mil22, Prop. 23.59]. Therefore, simply connected split semisimple groups are uniquely determined by their root systems.

3. PROOF OF THE MAIN THEOREM

Throughout this section, we use k to denote an number field. Fix an algebraic closure $\bar{k}$ of k , and let $k_s \subseteq \bar{k}$ denote the separable closure of k . We prove Theorem 1.2 by considering, one by one, the possible Dynkin types of the absolute root system of G . By results in Section 2, for each root system, there exists, up to k -isomorphism, a unique simply connected k -split semisimple group having that root system. Any simply connected semisimple group over k with the same absolute root system becomes isomorphic to this split group over k_s ; equivalently, it is a k -form of that group.

3.1. Type A_n .

3.1.1. *Inner Forms.*

First, assume that, after passing to the simply connected universal covering, G is an inner k -form of the simply connected k -split semisimple group with Dynkin diagram of type A_{n-1} . By [PR94, Prop. 2.17], G is isogenous to $\mathrm{SL}_m(D)$, where D is a central division algebra over k of degree n/m . Since G is assumed to be anisotropic, we must have $m = 1$, and hence $G \sim_{\mathrm{isog}} \mathrm{SL}_1(D)$. Now let N be a maximal subfield of D that is separable over k . Such a field exists by [Mil20, Prop. IV.3.8]. Then $[N : k] = n$ and N splits D . It is also known (see, e.g., [AT68, Ch. X]) that N may be chosen to be cyclic over k . Fix a prime divisor d of n . Since N/k is cyclic of degree n , there exists an intermediate field E with $[E : k] = n/d$. Let $B = C_D(E)$ be the centralizer of E in D . Then we know B is a central division algebra over E of degree d . Consequently, We may take the desired algebraic subgroup H to be $H = \mathrm{SL}_1(B)$.

Therefore, it remains only to verify the local assertion (3) of Theorem 1.2. Let v be a place of k . The central simple algebra $D_v := D \otimes_k k_v$ can be written as a matrix group $D_v \cong M_r(A_v)$ for some central division algebra A_v over k_v .

Since

$$G_{k_v} = G \times_k k_v \sim_{\text{isog}} \text{SL}_1(D_v) \cong \text{SL}_1(M_r(A_v)),$$

and G_{k_v} has k_v -rank 1 by assumption, we obtain

$$1 = \text{rank}_{k_v}(G_{k_v}) = \text{rank}_{k_v}(\text{SL}_1(M_r(A_v))) = r - 1.$$

Hence $r = 2$. In particular, $n = \deg(D_v) = 2 \deg(A_v)$ is even. We now try to construct an algebraic subgroup H with $d = 2$ satisfying the required local properties. By the previous construction, this is equivalent to finding an appropriate intermediate field E of degree $[E : k] = m = n/2$.

Suppose that $N_v := N \otimes_k k_v$ decomposes a product of two fields $N_v = N'_v \times N''_v$, each of which embeds into A_v . Assume moreover that $E_v := E \otimes_k k_v = E'_v \times E''_v$ also decomposes as a product of two fields (necessarily of the same degree and both embeds into A_v). In this case, the centralizer of E_v in D_v is isomorphic to $M_2(B_v)$, where B_v is the centralizer of E'_v in A_v . Therefore, H_{k_v} is isomorphic to $\text{SL}_2(B_v)$, and hence has k_v -rank 1.

It remains to show that there exists a pair (N, E) satisfying these conditions. This can be achieved by the weak approximation theorem. Since N/k is assumed to be cyclic, it corresponds to an Artin character $\epsilon : \text{Gal}(N/k) \rightarrow \mathbb{C}^\times$ of order n . The condition on N_v translates into the requirement that the local component ϵ_v has order $n/2$. Moreover, E corresponds to the character ϵ^2 , so the local component ϵ_v^2 has order $n/4$. It follows that E_v must decompose as a product of two field factors, each of degree $n/4$. Finally, by [AT68, Thm. X.6], we may choose ϵ (and hence N) so that the required conditions are also satisfied at every other ramified place of D .

3.1.2. Outer Forms.

We now assume that G is isogenous to an outer k -form of the simply connected k -split semisimple group with Dynkin diagram A_{n-1} . Up to central

isogeny, we may replace G by its universal covering, hence may assume that G is simply connected. In this case, [PR94, Prop. 2.18] together with our anisotropic assumption implies the existence of a quadratic extension E of k , a central division algebra D over E of degree $d = n$, and a D -module $V \cong D^\ell$, equipped with an anisotropic D -hermitian form h relative to an involution $*$ of the second kind on D , such that

$$G = \mathrm{SU}(D^\ell, h).$$

First assume that $d > 1$, with ℓ arbitrary. Let $v \in V$ be a nonzero vector. Since h is anisotropic, v is anisotropic, and its stabilizer $\mathrm{Stab}_G(v)$ is isomorphic to $\mathrm{SU}(W, h|_W)$, where $W = v^\perp \cong D^{\ell-1}$ is the orthogonal complement with respect to h . It follows by induction that G contains a group of type (1.2), the only issue being that d might not be prime. We will resolve this issue at the end of the section by choosing a prime divisor $p \mid d$, constructing a division algebra $B \subseteq D$ over E of degree p , and then considering the subgroup $H = \mathrm{SU}(B, *|_B) \subseteq \mathrm{SU}(D, *)$.

If, on the other hand, $d = 1$, then G is the special unitary group of an anisotropic hermitian form with respect to the quadratic extension E/k . If $\ell = 2$, then G itself is of type $\mathcal{A}$. If $\ell \geq 3$, then, by the same orthogonal-complement argument as above, G contains a subgroup corresponding to the case $\ell = 2$. Finally, if $\ell = 1$, then $G = \mathrm{SU}(E, h)$ is a norm-one torus, and hence is not semisimple. Thus, this case is excluded from consideration.

It remains to verify statement (3) of Theorem 1.2. We first consider the case $d = 1$. In this case we have $D = E$, and the involution $*$ of the second kind is simply the nontrivial k -automorphism of E . If the place v splits in E , then $E \otimes_k k_v \cong k_v \times k_v$, and $*$ induces the exchange involution $(a, b)^* = (b, a)$.

Accordingly, we obtain a decomposition

$$V \otimes_k k_v \cong (1, 0)(V \otimes_k k_v) \oplus (0, 1)(V \otimes_k k_v) := V_1 \oplus V_2.$$

Here V_1 and V_2 are k_v -vector spaces of dimension ℓ . The hermitian form h_v then satisfies $h_v(xa, yb) = a^*h_v(x, y)b$, and it follows that h_v vanishes on $V_1 \times V_1$ and $V_2 \times V_2$. Thus it induces a perfect pairing between V_1 and V_2 , identifying V_2 with the dual space $V_1^\vee$. Now take a unitary automorphism $g = (g_1, g_2) \in \mathrm{U}(V \otimes_k k_v, h_v)$ with $g_i \in \mathrm{GL}(V_i)$ for $i = 1, 2$. The unitary condition $h_v(gx, gy) = h_v(x, y)$ forces $g_2 = (g_1^{-1})^\vee$, so g is completely determined by g_1 . Hence we obtain an isomorphism $\mathrm{U}(V \otimes_k k_v, h_v) \cong \mathrm{GL}(V_1) \cong \mathrm{GL}_\ell(k_v)$. Under this identification, the special unitary condition becomes $\det(g_1) = 1$. Therefore,

$$G_{k_v} = \mathrm{SU}(V \otimes_k k_v, h_v) \cong \mathrm{SL}_\ell(k_v).$$

Since G has k_v -rank 1, it follows that $\ell = 2$. So G is of type $\mathcal{A}$.

If v is inert in E , then $(V \otimes_k k_v, h \otimes_k k_v)$ is a hermitian space over the quadratic extension $E \otimes_k k_v/k_v$. By [Sch85, Cor. 1.5.11], it admits a Witt decomposition

$$V \otimes_k k_v = \bigoplus_{i=1}^r P_i \oplus W,$$

where W is anisotropic, r is the Witt index, and the P_i are hyperbolic planes. Choose a Witt basis $\{e_1, \dots, e_r, f_1, \dots, f_r\}$ for the hyperbolic part such that $h(e_i, f_j) = \delta_{ij}$ and $h(e_i, e_j) = h(f_i, f_j) = 0$. With respect to this basis $G_{k_v} = \mathrm{SU}(V \otimes_k k_v, h \otimes_k k_v)$ contains a k_v -split torus of dimension r , given by

$$(t_1, \dots, t_r) \mapsto \mathrm{diag}(t_1, \dots, t_r, t_1^{-1}, \dots, t_r^{-1}, 1, \dots, 1),$$

which acts trivially on the anisotropic part W . Since G has k_v -rank 1, we must have $r = 1$. Thus $V \otimes_k k_v = P \oplus W$.

Take a nonzero vector $w \in W$. Then $\text{Stab}_{G_{k_v}}(w)$ is a subgroup of G_{k_v} of k_v -rank 1 and absolute rank $\ell - 1$. Since V is dense in $V \otimes_k k_v$ for the v -adic topology, we may approximate w by vectors $w_\alpha \in V$. For w_α sufficiently close to w , the restriction of h to the orthogonal complement $w_\alpha^\perp$ defines a hermitian form h' whose completion h'_v is similar to the restriction of $h \otimes_k k_v$ to $w^\perp$. Consequently,

$$\text{Stab}_G(w_\alpha) \times_k k_v \cong \text{Stab}_{G_{k_v}}(w).$$

Inducting on ℓ until $\ell = 2$, we conclude that G contains a subgroup H of type $\mathcal{A}$ such that H_{k_v} has rank 1.

We now assume that $d > 1$ and v is inert in E . By [Sch, Thm. 10.2.2], we have $D_v \cong M_d(E_v)$, so G_{k_v} is a genuine unitary group associated with an E_v/k_v -linear involution on $M_{d\ell}(E_v)$.

If v is non-archimedean, then Tits' table [Tit79] shows that the integer $d\ell$ can only be 2, 3, or 4. If $d\ell = 2$ or 3, then necessarily $\ell = 1$, and hence G is of type $\mathcal{A}$. The case $d = 4$ and $\ell = 1$ will be treated later, at the end of this section. Therefore, the only case left is $d = \ell = 2$.

In this case, by choosing an orthogonal basis of V , we may decompose h as the orthogonal sum of two hermitian forms h_1, h_2 on D -modules of rank 1. For each $i = 1, 2$, the scalar extension $h_i \otimes_k k_v$ is a hermitian form on $M_2(E \otimes_k k_v)$. Such a form may be represented by a matrix f_i satisfying $f_i = f_i^*$, and hence, by [Sch85, pp. 374], may be identified with a hermitian form on $(E \otimes_k k_v)^2$. Since G_{k_v} has rank 1, these two forms cannot be similar;; otherwise, their orthogonal sum would be hyperbolic, so that $(E \otimes_k k_v)^4$ would decompose as a direct sum of hyperbolic planes. It therefore has Witt index 2, a contradiction. Consequently, one of the two forms must be anisotropic, and the other isotropic. It follows that G contains a subgroup of the form $\text{SU}(D, *)$ of type $\mathcal{A}$ that is isotropic at v . This completes the proof in this case.

Next, suppose that v is a real place. Then $k_v \cong \mathbb{R}$ and $E_v \cong \mathbb{C}$. After base change to k_v , the involution of the second kind becomes ordinary complex conjugation on $\mathbb{C}$, and $h_v := h \otimes_k k_v$ is an ordinary hermitian form over $\mathbb{C}$. Writing $n = d\ell$, we therefore have $G_{k_v} \cong \mathrm{SU}_{p,q}$ for some integers $p + q = n$. Since G has real rank 1, we must have $\min(p, q) = 1$, hence h_v has signature $(1, n-1)$ up to ordering. By the same construction as in the previous paragraph, we may decompose

$$h = h_1 \perp \cdots \perp h_\ell,$$

where each h_i is a hermitian form on a rank-1 D -submodule of V . Equivalently, each h_i is associated with a symmetric element $f_i \in D^\times$. After extending scalars to k_v , the element f_i defines a hermitian form on $\mathbb{C}^d$. Since the orthogonal sum of these forms has signature $(1, n-1)$, one of them must have signature $(1, d-1)$. It follows that G contains a special unitary group H , associated with the corresponding rank-1 D -submodule, of the desired type.

We now turn to the most difficult case, namely $d > 1$ and v split in E . Since v splits, there are two places w and w' of E lying above v , and $E \otimes_k k_v \cong E_w \times E_{w'}$. After base change, we therefore obtain

$$D_v = D \otimes_k k_v \cong D \otimes_E (E \otimes_k k_v) \cong (D \otimes_E E_w) \times (D \otimes_E E_{w'}).$$

Write $D_w := D \otimes_E E_w$. Then the second factor $D_{w'}$ is naturally identified with the opposite algebra D_w^{op} , and the involution of the second kind on D_v becomes the exchange involution $(x, y) \mapsto (y, x)$ on $D_w \times D_w^{\mathrm{op}}$. Accordingly, we may decompose

$$V \otimes_k k_v \cong V_w \times V_{w'},$$

where V_w is a D_w -module of rank ℓ , and $V_{w'}$ is the corresponding D_w^{op} -module. The hermitian form identifies $V_{w'} \cong V_w^\vee$, and a unitary automorphism is completely determined by an arbitrary D_w -linear automorphism of V_w ; the

action on the $V_{w'}$ is then forced by the unitary condition. Consequently, $U(D_v, h_v) \cong \mathrm{GL}_\ell(D_w)$, and imposing determinant 1 gives

$$G_{k_v} = \mathrm{SU}(D_v, h_v) \cong \mathrm{SL}_\ell(D \otimes_E E_w).$$

We know G_{k_v} has rank 1 if $\ell = 2$ and $D \otimes_E E_w$ is a division algebra. Another possibility is that $\ell = 1$ and $D_w \cong M_2(B_w)$, where B_w is a central division algebra over $E_w \cong k_v$. We will focus on the $\ell = 2$ case first, and return to the case $\ell = 1$ at the end of this section.

As above, since $\ell = 2$, the group G is defined by a diagonal hermitian form on D^2 , say $h = \langle f_1, f_2 \rangle$, with $f_i \in D^\times$ satisfying $f_i = f_i^*$. Let H denote the algebraic group $U(D, *)$ defined over k . By [PR94, Prop. 2.16] rank-one hermitian forms over $(D, *)$ are classified up to similarity by the Galois cohomology set $H^1(k, H)$.

Moreover, by [Rog90, § 3.4], the maximal k -tori of H admit a more explicit description. Let $F = F_0 E$ be a subfield of D of degree d over E , where F_0 is a field extension of degree d over k that is linearly disjoint from E over k . Assume further that F is stable under the involution $*$, that $*$ restrict to the identity on F_0 , and that its restriction to E is the nontrivial automorphism in $\mathrm{Gal}(E/k)$. Then the torus $T \subseteq H$ is identified with

$$T_F = \ker(N_{F/F_0} : \underline{F}^\times \rightarrow \underline{F_0}^\times).$$

Here $\underline{F}^\times := \mathrm{Res}_{F/k} \mathbb{G}_m$ and $\underline{F_0}^\times := \mathrm{Res}_{F_0/k} \mathbb{G}_m$ denote the corresponding k -tori. In particular, T_F depends only on F_0 : indeed, $F = F_0 E$ once F_0 is fixed. Since v splits in E , we have $T_F \times_k k_v \cong \underline{F_0}^\times \times_k k_v$.

Now fix a Borel pair (B, T) of the base-changed group $H_{\bar{k}}$, and let $\Delta = \Delta(H_{\bar{k}}, B, T)$ be the corresponding set of simple roots. The *Langlands dual group* of H , denoted by $\widehat{H}$, is the connected complex reductive algebraic group

whose based root datum is the dual datum

$$\Psi^\vee(H_{\bar{k}}, T) = (X^\vee, \Phi^\vee, X, \Phi, \Delta^\vee).$$

Take $\Gamma = \text{Gal}(\bar{k}/k)$. Since Γ acts on $H_{\bar{k}}$, it also acts on $B, T, \Psi(H_{\bar{k}}, T)$, and hence on the dual root datum $\Psi^\vee(H_{\bar{k}}, T)$. By Theorem 2.10 this induces a canonical action of Γ on $\widehat{H}$. We then define

$$A(H/k) = (\pi_0(Z(\widehat{H})^\Gamma))^D,$$

where $Z(\widehat{H})^\Gamma$ denotes subgroup of Γ -fixed points in the center $Z(\widehat{H})$, $\pi_0(\cdot)$ denotes the group of connected components, and $(\cdot)^D = \text{Hom}(\cdot, \mathbb{C}^\times)$ denotes the Pontryagin duality for finite abelian groups.

Similarly, for each place v of k , we define a group $A(H/k_v)$. By [Kot86], there is a natural map $H^1(k_v, H) \rightarrow A(H/k_v)$, which is an isomorphism when v is finite. There is also a natural “sum” map $\Sigma : \bigoplus_v A(H/k_v) \rightarrow A(H/k)$. Hence, by [Kot88], we obtain an exact sequence of pointed sets

$$0 \rightarrow H^1(k, H) \xrightarrow{j} \bigoplus_v H^1(k_v, H) \xrightarrow{\Sigma} A(H/k) \rightarrow 0.$$

Here the injectivity of j follows from [RP, Thm. 6.27], which states that the unitary group H satisfies the Hasse principle.

In our setting, $H = \text{U}(D, *)$, where E/k a quadratic extension and D is a central division algebra of degree d over E . After base change to $\bar{k}$, we have $E \otimes_k \bar{k} \cong \bar{k} \times \bar{k}$ and $D \otimes_k \bar{k} \cong M_d(\bar{k}) \times M_d(\bar{k})$, with the involution $*$ interchanging the two factors. Consequently,

$$H_{\bar{k}} = \text{U}(D, *)_{\bar{k}} \cong \{(g, h) \in \text{GL}_d(\bar{k}) \times \text{GL}_d(\bar{k}) : h = (g^{-1})^t\}.$$

Projection onto the first factor therefore induces an isomorphism $H_{\bar{k}} \cong \text{GL}_d$. It follows that the root datum of H is the same as that of GL_d . More precisely, for

the diagonal torus T , we have $X(H_{\bar{k}}, T) \cong X^\vee(H_{\bar{k}}, T) \cong \mathbb{Z}^d$, and $\Phi(H_{\bar{k}}, T) = \{e_i - e_j\}_{i \neq j}$. This root datum is self-dual, and the complex Langlands dual group should be $\widehat{H} \cong \mathrm{GL}_d(\mathbb{C})$, whose center is $Z(\widehat{H}) \cong \mathbb{C}^\times$. The group Γ acts on $Z(\widehat{H})$ through $\mathrm{Gal}(E/k)$, whose generator acts by $z \mapsto z^{-1}$. Therefore, $A(H/k) \cong \mu_2$. The same computation applies locally at any place v that is inert in E .

Lemma 3.1. *Let $\alpha \in H^1(k, H)$, and write $j(\alpha) = (\alpha_v)$. Let $T_F \subseteq H$ be a maximal torus that is anisotropic over every place v for which $\alpha_v \neq 0$. Denote by $f : T_F \rightarrow H$ the inclusion. Then there exists $\beta \in H^1(k, T_F)$ such that $\alpha = f(\beta)$.*

Assuming this lemma, we obtain the following result, which is of independent interest.

Proposition 3.2. *Under the preceding hypotheses, let $f_1, \dots, f_r$ be symmetric elements of $D^\times$ (i.e. $f_i = f_i^*$). Then there exists symmetric elements $g_i \in D$, commuting pairwise, and elements $x_i \in D^\times$ such that $f_i = x_i g_i x_i^*$ for all i .*

We first prove Proposition 3.2 by assuming Lemma 3.1. Recall that over the separable closure k_s , the algebra with involution $(D, *)$ becomes split. More precisely, $D \otimes_k k_s$ is isomorphic to a matrix algebra over $E \otimes_k k_s \cong k_s \times k_s$, and $*$ becomes the standard unitary involution attached to the quadratic extension E/k . Under this identification, we have $D \otimes_k k_s \cong M_d(k_s) \times M_d(k_s)$, and the involution becomes $(A, B)^* = (B^t, A^t)$. Hence each symmetric element $f_i \in D^\times$ may be written in the form $f_i = (A_i, A_i^t)$ for some $A_i \in \mathrm{GL}_d(k_s)$. Define $x_i = (A_i, I_d) \in (D \otimes_k k_s)^\times$. We see that $f_i = x_i x_i^*$. Now let $\Gamma_s := \mathrm{Gal}(k_s/k)$. For each i we define a 1-cocycle

$$c_i : \Gamma_s \rightarrow H(k_s) : \sigma \mapsto x_i^{-1} \sigma(x_i).$$

Its image indeed lies in $H(k_s)$, since

$$x_i^{-1}\sigma(x_i)\sigma(x_i)^*(x_i^{-1})^* = x_i^{-1}\sigma(f_i)(x_i^{-1})^* = x_i^{-1}f_i(x_i^{-1})^* = 1.$$

A direct computation also shows that $c_i(\sigma\tau) = c_i(\sigma)\sigma(c_i(\tau))$ for all $\sigma, \tau \in \Gamma_s$. So c_i is indeed a 1-cocycle. Let $\alpha_i \in H^1(k, H)$ denote the corresponding cohomology class.

If T_F is anisotropic over every place v such that $(\alpha_i)_v \neq 0$ for some $i = 1, \dots, r$, then Lemma 3.1 shows that each α_i lies in the image of $H^1(k, T_F) \rightarrow H^1(k, H)$. By definition T_F fits into a short exact sequence of k -tori

$$1 \rightarrow T_F \rightarrow \underline{F}^\times \xrightarrow{N_{F/F_0}} \underline{F_0}^\times \rightarrow 1.$$

By [Spr98, Prop. 12.3.4], this gives rise to an exact sequence of pointed sets

$$1 \rightarrow T_F(k) \rightarrow F^\times \xrightarrow{N_{F/F_0}} F_0^\times \xrightarrow{\delta} H^1(k, T_F) \rightarrow H^1(k, \underline{F}^\times).$$

Hilbert's Theorem 90 implies that $H^1(k, \underline{F}^\times) = 1$. Hence δ is surjective, and therefore $H^1(k, T_F) \cong F_0^\times / N_{F/F_0}(F^\times)$. It follows that each class in $H^1(k, T_F)$, and in particular each α_i , is represented by some element $g_i \in F_0^\times$. Because F_0 is fixed pointwise by the involution $*$, each g_i is symmetric. Moreover, $F \otimes_k E \cong F_0 \otimes_k (E \times E) \cong F \times F$. Thus after base change to E , the torus T_F becomes

$$\begin{aligned} (T_F)_E &\cong \ker(N_{F/F_0} \otimes_k E : \underline{F}_E^\times \rightarrow \underline{F_0}_E^\times) \\ &\cong \ker((\text{Res}_{F/E}\mathbb{G}_m)^2 \rightarrow \text{Res}_{F/E}\mathbb{G}_m : (x, y) \mapsto xy) \cong \text{Res}_{F/E}(\mathbb{G}_m). \end{aligned}$$

This gives a natural identification $T_F(E) \cong F^\times$. Since $F_0^\times \subseteq F^\times$, we may view each g_i as an E -point of T_F . In particular, those elements g_i commute with one another.

To prove Proposition 3.2 it therefore remains to verify that there exists a torus T_F with the required local anisotropy property. By weak approximation

for the derived group of H [PR94, § 7.3], we may choose a regular element $\gamma \in H(k)$ that is elliptic at all the places under consideration. Then the centralizer of $Z_H(\gamma)$ is a maximal k -torus, and it is anisotropic at each of those places. This gives the desired torus T_F . Note that in this process we may ignore those places v that split in E , since for such v we have $H^1(k_v, H) = 1$.

We now proof Lemma 3.1. First, we note that $\widehat{T_F} \cong (\mathbb{C}^\times)^d$ and $T_F \times_k E \cong \text{Res}_{F/E}(\mathbb{G}_m)$. If we write $\Gamma_E = \text{Gal}(\bar{k}/E)$, then the subgroup of Γ_E -invariant points in $\widehat{T_F}$ is $(\widehat{T_F})^{\Gamma_E} \cong \mathbb{C}^\times$, embedded diagonally in $(\mathbb{C}^\times)^d$. Moreover, we also know $(\widehat{T_F})^\Gamma \cong \mu_2$ because the full Γ -action factors through $\Gamma/\Gamma_E \cong \text{Gal}(E/k) \cong \mathbb{Z}/2\mathbb{Z}$. Therefore, the inclusion $T_F \hookrightarrow H$ induces a morphism between the corresponding Kottwitz exact sequences in [Kot86]:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^1(k, T_F) & \xrightarrow{j_T} & \bigoplus H^1(k_v, T_F) & \xrightarrow{\Sigma_T} & A(T_F/k) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & H^1(k, H) & \xrightarrow{j} & \bigoplus H^1(k_v, H) & \xrightarrow{\Sigma} & A(H/k) \longrightarrow 0 \end{array}$$

The top row is exact at $H^1(k, T_F)$ because of the Hasse norm theorem. Furthermore, by [Kot86, § 10], if $T_F \times_k k_v$ is anisotropic, then the natural map $H^1(k_v, T_F) \rightarrow H^1(k_v, H)$ is surjective. It then follows from the snake lemma that Lemma 3.1 holds.

We now continue to prove part (3) of Theorem 1.2 in the case $d > 1$, $\ell = 2$, and v split in E . In this setting, G is associated with a hermitian form on D^2 . We may assume that the hermitian form is diagonal. By Lemma 3.1, we may further assume that $f_1, f_2 \in F_0^\times$, with $F = F_0E \subseteq D$. Set $H = \begin{pmatrix} f_1 & 0 \\ 0 & f_2 \end{pmatrix}$.

If $g \in \text{GL}_2(D)$ is written as $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d \in F$, then a direct

computation shows that

$$H^{-1}g^{*t}H = \begin{pmatrix} a^* & f_1^{-1}f_2c^* \\ f_2^{-1}f_1b^* & d^* \end{pmatrix}.$$

Let H_1 be the subgroup of $\mathrm{GL}_2(F)$ defined by the relation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} a^* & f_1^{-1}f_2c^* \\ f_2^{-1}f_1b^* & d^* \end{pmatrix}.$$

This is precisely the unitary group $\mathrm{U}(h)$ attached to the hermitian form $h(x, y) = x^*Hy$ on F^2 . After base-change to an algebraic closure $\overline{F_0}$, the quadratic extension F/F_0 splits as $F \otimes_{F_0} \overline{F_0} \cong \overline{F_0} \times \overline{F_0}$, and the involution is identified with the transposition of the two factors. Consequently, after this base change, the algebraic group H_1 becomes isomorphic to $\mathrm{GL}_{2, \overline{F_0}}$. In particular, H_1 has absolute rank 2.

Now recall that $D \otimes_E E_w$ is a division algebra. It follows that $F_0 \otimes_k k_v$ is a field. Since v splits in E and $F = F_0E$, the quadratic extension F/F_0 splits after base change to $F_0 \otimes_k k_v$:

$$F \otimes_{F_0} (F_0 \otimes_k k_v) \cong (F_0 \otimes_k k_v) \times (F_0 \otimes_k k_v).$$

Under this identification, the involution is given by exchanging the two factors. Therefore, after a suitable choice of basis, the defining equation $g^{*t}Hg = H$ takes the standard split form, and hence $H_1 \times_{F_0} (F_0 \otimes_k k_v) \cong \mathrm{GL}_{2, F_0 \otimes_k k_v}$. Now define $H_2 = \mathrm{Res}_{F_0/k} H_1$. Then H_2 is a k -subgroup of G , and

$$H_2 \times_k k_v \cong \mathrm{Res}_{F_0 \otimes_k k_v/k_v} (H_1 \times_{F_0} (F_0 \otimes_k k_v)) \cong \mathrm{Res}_{F_0 \otimes_k k_v/k_v} \mathrm{GL}_2.$$

In particular, H_2 has semisimple rank 1. Hence the derived group of H_2 is of type $\mathcal{A}$, and satisfies the condition in Theorem 1.2 (3).

We now turn to the remaining special cases that were left aside in the previous analysis. All those cases have the assumption $\ell = 1$. Thus

$$G = \mathrm{SU}(D^\ell, h) = \mathrm{SU}(D, *),$$

for some division algebra D over E of degree d . Let $G' = \mathrm{U}(D, *)$ denote the ambient group, and let p be a prime divisor of d . We shall use the description of the maximal tori of G' given above: these are the tori T_F , where F satisfies the necessary conditions for T_F to embed into G' .

Lemma 3.3 ([Kot92, pp. 421–422]). *Suppose that there exists a finite place v_0 that splits in E and is inert in F_0 . Then, if T_{F, k_v} embeds into G'_{k_v} for every place v of k , then T_F embeds into G' .*

Let us now determine under what conditions (of F_0) the local hypothesis of Lemma 3.3 is satisfied. We begin with the finite places. Suppose that v splits in E , and let w be a place of E lying above v . By the same argument as in the previous paragraphs, we have $G'_{k_v} \cong \mathrm{GL}_1(D_w)$ over k_v , where D_w is defined to be $D_w = D \otimes_E E_w$. Write $F_{0,v} := F_0 \otimes_k k_v$. Then giving an embedding $T_{F, k_v} = \mathrm{Res}_{F_{0,v}/k_v} \mathbb{G}_m \hookrightarrow G'_{k_v}$ is equivalent to giving a k_v -algebra embedding $F_{0,v} \hookrightarrow D_w$. Now suppose that v is inert in E , with $w \mid v$. By [Sch85, Thm. 10.2.2] we have $D_w \cong M_d(E_w)$. Moreover, involutions of the second kind on $M_d(E_w)$ are precisely the adjoint involutions attached to nondegenerate hermitian forms on E_w^d relative to the quadratic extension E_w/k_v . Hence there exists a hermitian form h on E_w^d such that

$$G'_{k_v} \cong \mathrm{U}(D_w, *) \cong \mathrm{U}(M_d(E_w), *) \cong \mathrm{U}(\mathrm{End}_{E_w}(E_w^d), *_h) \cong \mathrm{U}(E_w^d, h),$$

where $*_h$ denotes the adjoint involution defined by h . By [Mín11, § 3.2.1], if d is odd, then there is only one isomorphism class of d -dimensional unitary groups over E_w/k_v , which is quasi-split. It follows that G'_{k_v} is quasi-split. If d

is even, then there are two isomorphism classes of such unitary groups, one of which is quasi-split.

Write $F_v := F \otimes_k k_v$. Since v is inert in E , we also have $E_w \cong E \otimes_k k_v$. The field trace Tr_{F_v/E_w} defines a hermitian form $h_{\text{tr}} : F_v \times F_v \rightarrow E_w$ by $h(x, y) := \text{Tr}_{F_v/E_w}(xy^*)$. Then every element

$$a \in T_{F,v}(k_v) = \{a \in F_v^\times : aa^* = 1\}$$

satisfies $h_{\text{tr}}(x, y) = h_{\text{tr}}(ax, ay)$ for all $x, y \in F_v$. Therefore, each k_v -point a defines an E_w -linear automorphism of F_v preserving h_{tr} , and hence induces an embedding $T_{F,v} \hookrightarrow \text{U}(F_v, h_{\text{tr}})$. If d is odd, then G'_{k_v} is quasi-split, so $T_{F,v}$ embeds into G'_{k_v} . If d is even, we use that every maximal torus of the non-quasi-split unitary group also embeds into the quasi-split one. Hence $T_{F,v}$ embeds into G'_{k_v} whenever G'_{k_v} is quasi-split.

Finally, suppose G'_{k_v} is not quasi-split. Then G'_{k_v} is the unitary group of a hermitian form h on $V := E_w^d$, and V admits an orthogonal decomposition $V = V_2 \oplus V_{d-2}$, where V_2 is anisotropic and V_{d-2} is hyperbolic. Assume that $F_{0,v}$ can also be decomposed as $F_{0,v} = F_{v,2} \times F_{v,d-2}$ where $F_{v,2}$ is a quadratic field extension of k_v , linearly disjoint from E_w . Since $F = F_0 E$, we obtain

$$F_v \cong (F_0 \otimes_k k_v) \otimes_{k_v} E_v \cong (F_{v,2} \otimes_{k_v} E_v) \times (F_{v,d-2} \otimes_{k_v} E_v).$$

It follows that the torus can be decomposed as $T_F \otimes_k k_v \cong T_2 \times T_{d-2}$, where

$$T_2 = \ker(N : \underline{(F_{v,2} \otimes_{k_v} E_v)}^\times \rightarrow \underline{F_{v,2}}^\times),$$

$$T_{d-2} = \ker(N : \underline{(F_{v,d-2} \otimes_{k_v} E_v)}^\times \rightarrow \underline{F_{v,d-2}}^\times).$$

Here T_2 is anisotropic. Therefore T_2 embeds into $\text{U}(V_2)$, since anisotropic tori embed into every inner form; see [Kot86]. On the other hand, T_{d-2} embeds into $\text{U}(V_{d-2})$, because $\text{U}(V_{d-2})$ is quasi-split.

It remains to consider the archimedean places. Suppose first that v splits in E , and let w be a place of E above v . As before, we have $G'_{k_v} \cong \mathrm{GL}_1(D_w)$. If v is complex, then $k_v \cong \mathbb{C}$. Thus, to give an embedding $T_{F,k_v} \hookrightarrow G'_{k_v}$ is equivalent to giving a $\mathbb{C}$ -algebra embedding $F_0 \otimes_k \mathbb{C} \cong \mathbb{C}^d \hookrightarrow D_w \cong M_d(\mathbb{C})$, and such an embedding always exists. If otherwise v is a real place, then we seek an embedding $F_0 \otimes_k \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \hookrightarrow D_w$, where r_1 and r_2 denote the numbers of real and complex places of F_0 lying above v , respectively. By Frobenius' theorem, the Brauer group of $\mathbb{R}$ is $\mathrm{Br}(\mathbb{R}) = \{[\mathbb{R}], [\mathbb{H}]\} \cong \mathbb{Z}/2\mathbb{Z}$, where $\mathbb{H}$ denotes the Hamilton quaternion algebra. Since D_w has degree d , we must have either $D_w \cong M_d(\mathbb{R})$ or $D_w \cong M_{d/2}(\mathbb{H})$. In the first case, the embedding $F_0 \otimes_k \mathbb{R} \hookrightarrow D_w$ exists automatically. In the second case such an embedding exists if and only if F_0 is totally imaginary above v .

Now suppose that v is inert in E . Then $k_v \cong \mathbb{R}$ and $E_w \cong E \otimes_k \mathbb{R} \cong \mathbb{C}$. In this case, the torus $T_{F,v} := T_F \times_k k_v$ has real points

$$T_{F,v}(\mathbb{R}) \cong (\mathbb{C}^\times)^{r_2} \times T_c^{r_1},$$

where T_c denotes the anisotropic one-dimensional torus over $\mathbb{R}$. We also know that $G'_{k_v} \cong \mathrm{U}(\mathbb{C}^d, h) = \mathrm{U}(p_v, q_v)$ for suitable integers $p_v + q_v = d$. After permuting p_v and q_v if necessary, we may assume that $p_v \geq q_v$. The local condition at v is then $r_2 \leq q_v$.

We may now summarize the preceding discussion in the following statement.

Lemma 3.4. *Let F_0/k be an extension of degree d that is linearly disjoint from E . Assume that F_0 satisfies the following conditions:*

- (1) *For every split place v , the algebra $F_{0,v}$ embeds into D_w .*
- (2) *For every finite inert place v such that G'_{k_v} is not quasi-split, we have $F_{0,v} \cong F_{v,2} \times F_{v,d-2}$ where $F_{v,2}$ is a quadratic extension of k_v linearly disjoint from E_w .*
- (3) *For every archimedean inert place v , we have $r_2(v) \leq q_v$.*

(4) *There exists a finite place v_0 which is split in E and is inert in F_0 .*

Then T_F embeds into G' .

These conditions on F_0 involve only finitely many places of k . By weak approximation, we can arrange them for a suitable extension F_0/k . (Note that condition (4) already implies that F_0 is linearly disjoint from E .) It therefore remains to show that one may choose F_0 so that it contains a subfield M_0 , linearly disjoint from E , of degree d/p . Let us first reformulate condition (1) of Lemma 3.4. Let v be a place of k that splits in E , and w be a place of E above v . By [Jac89, § 9.14], the local field E_w has Brauer group $\text{Br}(E_w) \cong \mathbb{Q}/\mathbb{Z}$. Since D_w has degree d , its Brauer class has finite order r for some divisor $r \mid d$. Accordingly, we may write $D_w \cong M_{d/r}(B_r)$, where B_r is a division algebra over E_w . Recall that if w' denotes the other place of E above v , then $D_{w'}$ is identified with the opposite algebra D_w^{op} . Hence its Brauer class satisfies $[D_{w'}] = -[D_w]$, implying that they have the same order. Therefore, $r = r(v)$ only depends on the choice of v , not on w . The condition (1) is hence equivalent to

$$(3.1) \quad \begin{array}{l} \text{For every split place } v \text{ such that } D_w \text{ is ramified,} \\ F_{0,v} \text{ is a product of fields whose degree are divisible by } r(v). \end{array}$$

Suppose that v splits in E and that D_w is ramified. With $r(v)$ defined as above, write $r(v) = r_1(v)p^\alpha$, where $\alpha \in \{0, 1\}$ and $r_1(v) \mid (d/p)$. By weak approximation, we may choose an extension M_0/k of degree d/p such that, for every such place v , $M_{0,v} := M_0 \otimes_k k_v$ is a product of fields whose degrees over k_v are divisible by $r_1(v)$, and such that condition (2) and (4) of Lemma 3.4 are satisfied. We also assume that every real place of k that is inert in E splits as a product of real places in M_0 . We then choose F_0/M_0 of degree p satisfying Lemma 3.4. More precisely, if v is split over E and D_w is ramified, then for

each place of M_0 above v we require F_0/M_0 to split completely when $\alpha = 0$, and to remain of degree p when $\alpha = 1$.

Set $F = F_0E$ and $M = M_0E$. Then the torus T_F embeds into G' . Moreover, T_F contains T_M , whose k -points are given by

$$T_M(k) = \{x \in M^\times : xx^* = 1\}.$$

Viewing $T_M(k)$ as a subset of $M^\times \subseteq M$, we claim that $T_M(k)$ contains a primitive element of M/k . To prove this claim, it suffices to show that $T_M(k)$ is not contained in the finite union of the proper intermediate fields of M/k . By Hilbert's Theorem 90, we know that $H^1(\text{Gal}(M/M_0), M^\times) = 1$. Equivalently, every element $x \in M^\times$ with $N_{M/M_0}(x) = xx^* = 1$ can be written in the form $x = y/y^*$ for some $y \in M^\times$. Thus the map

$$M^\times \rightarrow T_M(M_0) : \quad y \mapsto \frac{y}{y^*}$$

induces an isomorphism $M^\times/M_0^\times \cong T_M(M_0)$. Since $M^\times/M_0^\times$ is birational to the projective space over M_0 , it follows that T_M is a rational variety of dimension 1 over M_0 . This, together with the fact that T_M is irreducible and k as a number field is infinite, implies that $T_M(k)$ is Zariski dense in T_M .

Now argue by contradiction. Suppose that $T_M(k)$ is contained in a finite union $\bigcup_{i=1}^r L_i$ of proper intermediate fields L_i of M/k . Each L_i is a k -linear subspace of M . Regard M as an affine k -space by $M \cong \mathbb{A}_k^{2d/p}$, we see L_i is Zariski closed in M . Since T_M is irreducible and $T_M(k)$ is Zariski dense in T_M , it would follow that $T_M \subseteq L_i$ for some i .

We now show that this cannot occur. After base change to $\bar{k}$, we have $M \otimes_k \bar{k} \cong \bar{k}^{2d/p}$. After reordering the coordinates, we may assume that the involution $*$ exchanges the coordinates in pairs:

$$(x_1, x_2, \dots, x_{2d/p-1}, x_{2d/p})^* = (x_2, x_1, \dots, x_{2d/p}, x_{2d/p-1}).$$

Accordingly, $T_M \times_k \bar{k}$ identifies with the subset of $\mathbb{A}_{\bar{k}}^{2d/p}$ defined by $x_{2j-1}x_{2j} = 1$ for $1 \leq j \leq d/p$. Now let $\sum_{i=1}^{2d/p} \lambda_i x_i = 0$ be a linear equation vanishing on all of $T_M(\bar{k})$. Substituting $(x_1, \dots, x_{2d/p}) = (t_1, t_1^{-1}, \dots, t_{d/p}, t_{d/p}^{-1})$ we obtain

$$\sum_{j=1}^{d/p} (\lambda_{2j-1} t_j + \lambda_{2j} t_j^{-1}) = 0$$

for all $t_1, \dots, t_{d/p} \in \bar{k}^\times$. Since $\bar{k}$ is infinite, it follows that all coefficients λ_i must be zero. Thus no nonzero linear form vanishes identically on $T_M(\bar{k})$. Equivalently, $T_M \times_k \bar{k}$ is not contained in any proper linear subspace of $\bar{k}^{2d/p}$.

Therefore, we may choose a primitive element $x \in T_M(k)$. The centralizer of x in D is then a division subalgebra B of degree p over M , stable under the involution $*$. Consequently, $*$ induces on B an involution of the second kind relative to M/M_0 . It follows that the group $H := \mathrm{SU}(B, *)$ is of type $\mathcal{A}$ contained in $G = \mathrm{SU}(D, *)$.

It remains to verify the local conditions, that is, Part (3) of Theorem 1.2. Let v be a finite place of k . Suppose that v splits in E , and let $w \mid v$. Then $D_w := D \otimes_E E_w$ must be isomorphic to $M_p(B_w)$ for a division algebra B_w over $E_w \cong k_v$. In particular, in the present situation we must have $p = 2$. The field $M_v := M \otimes_k k_v$ is of degree $d/2$ over k_v . It is therefore enough to impose at v on M_0 and F_0 the decomposition conditions explained at the end of Section 3.1.2.

Now suppose that v is inert in E . We have already shown that G_{k_v} is a genuine unitary group, whose rank can be equal to 1 only if $d = 2, 3$, or 4. If $d = 4$, then G_{k_v} is not quasi-split, M_0 is quadratic over k , and by assumption $M_{0,v} := M_0 \otimes_k k_v$ is a quadratic extension of k_v linearly disjoint from $E_v := E \otimes_k k_v$. The center of $\mathrm{U}(B, *) (k_v)$, which is isomorphic to

$$T_M(k_v) = \ker (N : (M_{0,v} \otimes F_v)^\times \rightarrow M_{0,v}^\times)$$

is therefore anisotropic. On the other hand,

$$T_F(k_v) \cong \ker \left(N : (F_{0,v} \times F_v)^\times \rightarrow F_{0,v}^\times \right),$$

where $F_{0,v}$ satisfies condition (2) of Lemma 3.4. Its k_v -rank is 1 if, as we may assume $F_{v,d-2}$ (here $d-2=2$) is isomorphic to E_v . In this case H_{k_v} is therefore isotropic, as required.

Finally, let v be an archimedean place. We may assume that $d \geq 3$ (otherwise $d=p$) and that v is real. If v is split, then necessarily $d=4$, $p=2$, and $D_w \cong M_2(\mathbb{H})$. We assume that $M_{0,v} \cong \mathbb{R} \times \mathbb{R}$, in which case necessarily $F_{0,v} \cong \mathbb{C} \times \mathbb{C}$. The centralizer of $M_{0,v}$ in D_w is then isomorphic to $M_2(\mathbb{R})$.

Now suppose that v is inert. With the notation introduced before Lemma 3.4, we have $q_v = 1$. We assume that M_0 is totally real. By an argument analogous to the one used in the p -adic case, the center of $U(B,*)(k_v)$ is anisotropic. If we further assume that $r_2 = 1$ in the sense of Lemma 3.4, then $T_F(k_v)$ has rank 1. Consequently, H_v also has rank 1.

3.2. Type B_n .

Assume G is of type B_n , and let $T \subseteq G$ be a maximal k -torus. Denote by $\Phi(G, T)$ the set of roots in the absolute root datum $\Psi(G_{k_s}, T_{k_s})$. Recall that, for the root system B_n , the set of long roots forms a closed subsystem of type D_n . It follows from [MT11, Cor. 13.7], the one-parameter subgroups associated with these long roots generate a k -subgroup $H \subseteq G$ of type D_n containing T . Thus the argument reduces to the case of type D_n . If G has k_v -rank 1, we may choose T so that $T \times_k k_v$ has rank 1. Therefore, the local hypothesis at v is also satisfied for H .

We also recall the criterion stated at the end of Section 2: G is simply connected if and only if $X(G, T)$ coincides with the weight lattice $P(\Phi(G, T))$.

Equivalently, the coroots span the entire cocharacter lattice $X^\vee(G, T)$. Applying this criterion to the coroot systems of types B_n and D_n , we can conclude that G is simply connected then so is H .

3.3. Type C_n .

In this case, by [CU04] we know the unipotent subgroups associated with the long roots generates a subgroup of G which is, up to isogeny, an anisotropic form of SL_2^n over k , and therefore, contains a subgroup of type (1.1), modulo restriction of scalars. The argument concerning the rank and simply connectedness is similar to the case B_n .

3.4. Type D_n .

Recall that in general algebraic group theory, a group G of classical type is further said to be *classical* if it is not of the type 3D_4 or 6D_4 according to Tits' table [Tit79]. Those are two explicit outer forms with Dynkin diagram D_4 , called *trialitarian groups*. More explicit definition for those can be found in [Spr98, § 17.3].

3.4.1. Classical Groups.

We first assume that G is classical. Up to isogeny we may assume G is simply connected. Then (see, e.g., Case (2) and (5) in [PR94, pp. 92]) G must fall in one of the following two descriptions.

First, it might be the case that $G = \mathrm{Spin}_{2n}(f)$ is the spin group of a quadratic form f . It is the simply connected universal covering for the orthogonal group $G' = \mathrm{SO}_{2n}(f)$. Since $n \geq 4$, G' contains an orthogonal group H in 4 variables. Its universal covering $\tilde{H} \subseteq G$, after base change to $\bar{k}$, is isomorphic to Spin_4 , with root system $A_1 \times A_1$. The type A_1 has no outer forms, so every simply connected k -form of A_1 is an inner form of SL_2 , and therefore is of the form $\mathrm{SL}_1(D)$ for some central simple algebra D of degree 2 over k . Such D is either $M_2(k)$ or a quaternion division algebra. In either case this is of type

(1.1). If G_{k_v} has rank 1, then we can choose such a group H to be isotropic at v , thus completing the rank requirement.

Otherwise, G is the universal covering of $\mathrm{SU}(B^n, f)$, where B is a quaternion algebra and f is a form on B^n , skew-hermitian with respect to the canonical involution of B . By [PR94, Prop. 2.21] we pick an anisotropic vector $a \in B^n$, then the stabilizer group $\mathrm{Stab}_G(a) \subseteq G$ is the universal covering of

$$\mathrm{SU}((Ba)^\perp, f|_{(Ba)^\perp}) \cong \mathrm{SU}(B^{n-1}, f')$$

for some appropriate form f' . This process can be recurrently invoked as long as $n \geq 3$. So in the end we can pick a subgroup $H \subseteq G$ which is the universal covering of a group of form namely $\mathrm{SU}(B^2, f')$ for a suitable skew-hermitian form f' on B^2 . It follows that H is a k -form of a semisimple group with root system $A_1 \times A_1$. By identical argument in the preceding paragraph we know H is k -isogenous to a product of two groups of type (1.1).

It remains to show that in this way we can construct a subgroup H that is isotropic at v when $G \times_k k_v$ has rank 1. Let $V = B^n$, viewed as a hermitian space for the form f . If G has rank 1, then by [Sch85, § 7.7] we have

$$V \otimes_k k_v \cong P \oplus W,$$

where P is a hyperbolic plane and W is anisotropic. By [Sch85, Lem. 7.6.2] we can pick $w \in W$ such that $f(w, w) \neq 0$. Then we can write w as a limit, for the v -adic topology, of a sequence $w_\alpha \in V$. For α sufficiently large we have $f(w_\alpha, w_\alpha) \neq 0$. The group G then contains the subgroup $H_\alpha := \mathrm{Stab}_G(w_\alpha)$. By induction on dimension, it suffices to show that $H_\alpha \times_k k_v$ is isotropic for α sufficiently large.

Let $\mathfrak{g} = \mathrm{Lie}(G \times_k k_v)$, and let $\mathfrak{h}_\alpha \subseteq \mathfrak{g}$ be the Lie algebra of the derived group of H_α . If $N = \dim(\mathfrak{h}_\alpha)$, we may regard $\mathfrak{h}_\alpha$ as a point of the Grassmannian X

of N -planes in $\mathfrak{g}$. The subset $Y \subseteq X$ consists of semisimple Lie subalgebras is a quasi-projective variety defined over k_v .

Moreover, $\mathrm{Lie}(\mathrm{Stab}_G(w_\alpha))$ tends to $\mathrm{Lie}(\mathrm{Stab}_G(w))$ in the appropriate Grassmannian, which implies that the derived Lie algebra $\mathfrak{h}$ of $\mathrm{Lie}(\mathrm{Stab}_G(w))$ is contained in the limit of the $\mathfrak{h}_\alpha$'s. Since they have the same dimension, we know $\mathfrak{h} = \lim(\mathfrak{h}_\alpha)$. Now the set of elements of $Y(k_v)$ consisting of the subalgebras isomorphic to $\mathfrak{h}$ is closed. This fact follows from the compactness of the projective space $\mathbb{P}(\mathrm{Hom}_{\mathrm{Lie}}(\mathfrak{h}, \mathfrak{g}))$ and from the semisimplicity of $\mathfrak{h}$. It is also open, since there are only finitely many semisimple Lie algebras over k_v of a given dimension. For α sufficiently large, $\mathfrak{h}_\alpha$ is therefore isomorphic to $\mathfrak{h}$. But $\mathrm{Stab}_G(w)$ contains the group of isometries of a hyperbolic plane, and is therefore isotropic. Now, $\mathfrak{h}_\alpha \cong \mathfrak{h}$ implies that the derived group of $H_\alpha \times_k k_v$ is of the same semisimple type as the derived group of $\mathrm{Stab}_G(w)$. So $H_\alpha \times_k k_v$ is also isotropic.

3.4.2. Trialitarian Groups.

It remains to deal with the groups of type 3D_4 and 6D_4 in the classification of Tits [Tit79]. These groups are studied by Harder [Har65], who shows that they satisfy the Hasse principle. If G is such a group, Harder shows that there exists a totally imaginary quadratic extension E of k with the following property. We can find a maximal torus $T \subseteq G$, defined over k , whose maximal subtorus S splits over E (a k -subtorus of T) is a maximal torus of a semisimple group $H_1 \subseteq G$, where H_1 is an outer k -form of the split group of type A_2 . Equivalently, H_2 is a unitary group [Har65, before Lem. 4.2.2.1].

We can write $H_1 \sim_{\mathrm{isog}} \mathrm{SU}(D^\ell, h)$ for a division algebra D over E , having degree d such that $d\ell = 3$. If D is a nontrivial division algebra, over E , then after base change to E the group becomes the inner form $\mathrm{SL}_1(D)$, which is not split. However, $S \subseteq H_1$ is split over E , which implies the group $H_1 \times_k E$

should be split. So the only possible case is $D = E$ and $H_1 \sim_{\text{isog}} \text{SU}(E^3, h)$, which is a genuine unitary group relative to E/k .

Since the hermitian form h must be non degenerate, there exists some vector $a \in E^3$ such that $h(a, a) \neq 0$. Take the subgroup $H \subseteq H_1$, which is isogenous to $\text{SU}(a^\perp, h|_{a^\perp})$. We know H is of absolute type A_1 . Type A_1 has no outer forms, so H is automatically of the form $H \cong \text{SL}_1(B)$, where B is a quaternion division algebra over k or $B = M_2(k)$. In either case H is in the form of (1.1). If v is a finite place, then $H_1 \times_k k_v$ is isomorphic to $\text{SL}_3(k_v)$ or $\text{SU}_3(k_v)$ — the unique unitary group of absolute rank 2, hence of rank 1 over k_v . If G_{k_v} has rank 1 (and therefore v is inert in E), we can choose H_{k_v} isotropic.

It remains to verify the real places. By Case (2) in [PR94, pp. 92] we know for a simply connected group of absolute type D_4 , the real forms are the simply connected covers of the special orthogonal groups attached to quadratic forms of dimension 8. More concretely, if we assume G to be simply connected, we must have $G \times_k k_v \cong \text{Spin}_{p,q}$ with $p + q = 8$. But the constraint on rank gives us $\text{rank}_{\mathbb{R}} \text{Spin}_{p,q} = \min(p, q) = 1$, leaving the only possibility

$$G \times_k k_v \cong \text{Spin}_{1,7}.$$

This group does not contain an anisotropic maximal torus over $k_v = \mathbb{R}$. That is because by [Mil22, pp. 253] a torus T over $\mathbb{R}$ is anisotropic if and only if $T(\mathbb{R})$ is compact. However, the maximal compact subgroup of G is Spin_7 , which has absolute rank 3; while a maximal torus of G should have absolute rank 4. Recall that the Galois group $\text{Gal}(\overline{k_v}/k_v) = \text{Gal}(\mathbb{C}/\mathbb{R})$ acts on the based root datum of G , inducing an action on its Dynkin diagram. If this action is trivial, then G is an inner form of the split group with type D_4 , implying that G contains an anisotropic maximal torus, leading to a contradiction. So the Galois group must act nontrivially, hence G must be of type 6D_4 over k .

Let K/k be the extension (with Galois group $\mathfrak{S}_3$) over which G becomes an inner form of a split group. Then by [Har65] (see also [PR94, Lem. 6.29]), there exists a totally imaginary quadratic extension E of k , linearly disjoint from K , over which G becomes quasi-split.

Let σ be a generator of $\text{Gal}(E/k)$. By [Har65] (see also [PR94, Lem. 6.17]), we may take B be a Borel subgroup of $G \times_k E$ such that $T = B \cap B^\sigma$ is a maximal torus of G defined over k , where B^σ denotes the Galois conjugate. By construction, K is the splitting field of the Dynkin-diagram action, and E is the quadratic extension involved in the outer form. Passing to the composite field EK kills both kinds of twisting. That means the base-change G_{EK} is a split semisimple group. As Borel subgroups of a split group, we know B_{EK} and B_{EK}^σ are also split. Write $R_u(B_{EK})$ to be the unipotent radical, we consider the composition map

$$T_{EK} \hookrightarrow B_{EK} \twoheadrightarrow B_{EK}/R_u(B_{EK}).$$

Its kernel $T_{EK} \cap R_u(B_{EK})$ is trivial because a torus has no nontrivial unipotent subgroup. Together with the maximality of T_{EK} we know this map is an isomorphism. Thus T splits over EK .

Recall that $\text{Gal}(\bar{k}/k)$ acts on the character group $X(T_{\bar{k}})$. But because T_{EK} is split, this action factors through the quotient $\text{Gal}(\bar{k}/k)/\text{Gal}(\bar{k}/EK) \cong \text{Gal}(EK/k) := \mathfrak{g}$. If we identify $\text{Gal}(E/k)$ with the subgroup $\mathfrak{g}_2 := \text{Gal}(EK/K)$ of $\mathfrak{g}$, and $\text{Gal}(K/k)$ with $\mathfrak{g}_6 = \text{Gal}(EK/E)$, then σ acts by (-1) on $X(T_{\bar{k}})$, whereas $\mathfrak{g}_6$ acts through the unique action determined by its action on the Dynkin diagram. Let $G_1 \cong \text{SL}_2/\bar{k}$ be the subgroup of G associated with the central root of that diagram, and let $G_2 \cong \text{SL}_2^3/\bar{k}$ be the subgroup associated with the three outer roots. Then G_1 and G_2 are defined over k , and $T = T_1 \times T_2$ where $T_i = T \cap G_i$ for $i = 1, 2$. Since T_{k_v} is isotropic, the same is true for T_1

or T_2 . The desired subgroup H is therefore G_1 or G_2 . Note that G_1 and G_2 are simply connected.

It remains to verify that in every case we may choose H simply connected whenever G is. This is clear in classical cases. For the groups associated with triality, this is true, as we have seen, for the construction of real place v . It is likely that the group H_1 defined by Harder is simply connected. (It is mentioned in [Clo03] that the author have checked the simply connectedness of H_1 , assuming it is connected. However, no explicit argument was given.) But in any case, since the representation of SL_2 in SL_3 with irreducible degrees $(2, 1)$ has simply connected image in PGL_3 , our H is also simply connected.

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